

Properties of Weighted Sobolev Spaces

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1. Weighted Sobolev Spaces

Let $n \geq 2$ be given and fix $1 \leq p < \infty$. Henceforth, Ω denotes an open subset of $\mathbb{R}^n$. A Lebesgue measurable function $\omega : \Omega \rightarrow (0, \infty]$ is called a (*lower Lebesgue*) *weight* provided each of the following criteria hold:

- (1) $\omega \in L^1_{\text{loc}}(\Omega)$,
- (2) for every compact set $K \subset \Omega$, there exists $c > 0$ such that $\omega(x) \geq c$ for almost every $x \in K$.

In addition, we will say that ω *dominates* the Lebesgue measure on Ω if there exists $c > 0$ with the property that $\omega(x) \geq c$ for almost every $x \in \Omega$, i.e. if $\omega \gtrsim 1$ on Ω . In this document, unless stated otherwise, ω will *always* be a lower Lebesgue weight.

DEFINITION 1. The weighted Sobolev space $W^{1,p}(\Omega, \omega)$ is defined to be the real vector space of all weakly differentiable functions $u \in L^p(\Omega, \omega)$ such that $\nabla u \in L^p(\Omega, \omega)$. Here, $L^p(\Omega, \omega)$ consists of those measurable functions f satisfying

$$\int_{\Omega} |f|^p \omega \, dx < \infty. \quad (1.1)$$

Or, equivalently,

$$\int_{\Omega} |f|^p \, d\mu < \infty. \quad (1.2)$$

with the measure μ given by $\mu(E) = \int_E \omega \, dx$. The space $W^{1,p}(\Omega, \omega)$ is equipped with the norm

$$\begin{aligned} \|u\|_{W^{1,p}(\Omega, \omega)} &:= \|u\|_{L^p(\Omega, \omega)} + \|\nabla u\|_{L^p(\Omega, \omega)} \\ &= \left(\int_{\Omega} |u|^p \omega \, dx \right)^{1/p} + \left(\int_{\Omega} |\nabla u|^p \omega \, dx \right)^{1/p}. \end{aligned}$$

Again, this norm can be equivalently described in terms of the measure μ . Finally, we denote $W_0^{1,p}(\Omega, \omega)$ the closure of $C_c^\infty(\Omega)$ in $W^{1,p}(\Omega, \omega)$.

We will soon see that $W^{1,p}(\Omega, \omega)$ is a Banach space, as is the case with the standard Sobolev spaces. However, until then, some observations are in order:

- Firstly, μ is automatically defined on all Lebesgue measurable sets E . Moreover, since ω is strictly positive on Ω , it is clear that μ and the Lebesgue measure have precisely the same null sets. Hence, both these measures induce the same notion of “almost everywhere”.
- Second, $\omega \in L^1_{\text{loc}}(\Omega)$ ensures that μ is finite on all compact subsets of Ω . If $K \subset \Omega$ is compact then

$$\mu(E) = \int_E \omega \, dx$$

is finite by the local integrability of ω .

- Finally, $u \in W^{1,p}(\Omega, \omega)$ must automatically belong to $L^1_{\text{loc}}(\Omega)$. Indeed, for any compact $K \subset \Omega$, we have $u \in L^p(K, \omega)$ so that

$$\int_K |u|^p \, d\mu < \infty$$

where $d\mu = \omega \, dx$ and $\mu(K) < \infty$ by the previous bullet. This ensures that $u \in L^1(K, \omega)$ whence

$$\int_K |u| \, dx \leq \frac{1}{c} \int_K |u| \omega \, dx = \frac{1}{c} \int_K |u| \, d\mu < \infty.$$

by property (2). It therefore does indeed make sense to discuss the weak derivatives of these u . In fact, since we only required that $u \in L^p(\Omega, \omega)$, we see that $L^p(\Omega, \omega) \subseteq L^1_{\text{loc}}(\Omega)$.

As promised, we now assert that $W^{1,p}(\Omega, \omega)$ is Banach whenever ω is a lower Lebesgue weight on Ω .

THEOREM 1.1. *Under the aforementioned conditions, $W^{1,p}(\Omega)$ and $W_0^{1,p}(\Omega)$ are both Banach spaces.*

PROOF. After replacing $\omega \, dx$ with $d\mu$, it is clear that $\|\cdot\|_{W^{1,p}(\Omega, \omega)}$ is a norm on $W^{1,p}(\Omega, \omega)$ and $W_0^{1,p}(\Omega, \omega)$. By definition of $W_0^{1,p}(\Omega, \omega)$, it will be enough to show that $W^{1,p}(\Omega, \omega)$ is complete. To this end, let (u_α) be a Cauchy sequence in $W^{1,p}(\Omega, \omega)$ so that

$$u_\alpha \rightarrow v \quad \text{and} \quad \partial_i u_\alpha \rightarrow v_i$$

in $L^p(\Omega, \omega)$, as $\alpha \rightarrow \infty$, for all $i = 1, \dots, n$. By our third observation, v and each v_i are locally integrable (with the Lebesgue measure) on Ω . Given $\varphi \in C_c^\infty(\Omega)$, we can find a compact set $K \subset \Omega$ such that $\text{supp}(\varphi) \subseteq K$. Utilizing once more property

(2), we have $u_\alpha \rightarrow v$ in $L^p_{\text{loc}}(\Omega)$ so that

$$\begin{aligned} \int_{\Omega} u \partial_i \varphi \, dx &= \lim_{\alpha \rightarrow \infty} \int_{\Omega} u_\alpha \partial_i \varphi \, dx = - \lim_{\alpha \rightarrow \infty} \int_{\Omega} \varphi \partial_i u_\alpha \, dx \\ &= - \int_{\Omega} \varphi v_i \, dx \end{aligned}$$

where we have used a similar argument in this last equality as well. It follows that $\nabla v = (v_1, \dots, v_n)$ and $u_\alpha \rightarrow v$ in $W^{1,p}(\Omega)$. This completes the proof. $\square$

1.1. Extension by Zero and Pointwise Results. We now turn towards the subspace $W_0^{1,p}(\Omega, \omega)$, which is slightly more convenient to work with. As with classical Sobolev spaces, $W_0^{1,p}(\Omega, \omega)$ should be interpreted as those functions $u \in W^{1,p}(\Omega, \omega)$ that vanish on $\partial\Omega$. This perspective is further motivated by the following extension theorem:

PROPOSITION 1.2. *Let V be an open set in $\mathbb{R}^n$ containing $\bar{\Omega}$ and let $u \in W_0^{1,p}(\Omega, \omega)$ be given. Assume that η is a (lower Lebesgue) weight on V such that $\eta(x) \leq C\omega(x)$ a.e. on Ω , for a suitable constant $C > 0$. After extending u by zero on $V \setminus \Omega$, we have $u \in W_0^{1,p}(V, \omega)$ and, moreover, $\|u\|_{W^{1,p}(\Omega, \eta)} = C \|u\|_{W^{1,p}(V, \omega)}$. In particular, there is an embedding*

$$W_0^{1,p}(\Omega, \omega) \hookrightarrow W_0^{1,p}(V, \eta)$$

obtained by sending u to $u\mathbf{1}_\Omega$.

PROOF. Choose a sequence (φ_α) in $C_c^\infty(\Omega)$ converging to u in $W_0^{1,p}(\Omega, \omega)$. In particular, the sequences (φ_α) and $(\nabla \varphi_\alpha)$ are Cauchy in $L^p(\Omega, \omega)$. Since each φ_α belongs to $C_c^\infty(V)$ by extension and has compact support in Ω , the sequence (φ_α) is also Cauchy in $W^{1,p}(V, \eta)$. To see this, simply note that for all $\alpha, \beta \in \mathbb{N}$ there holds

$$\|\varphi_\alpha - \varphi_\beta\|_{W^{1,p}(V, \eta)} = \|\varphi_\alpha - \varphi_\beta\|_{W^{1,p}(\Omega, \eta)} \leq C^{1/p} \|\varphi_\alpha - \varphi_\beta\|_{W^{1,p}(\Omega, \omega)}. \quad (1.3)$$

Thus, we can find $v \in W^{1,p}(V, \eta)$ such that $\varphi_\alpha \rightarrow v$ in $W^{1,p}(V, \eta)$. Passing to subsequences if necessary, we may assume without loss of generality that $\varphi_\alpha \rightarrow u$ a.e. on Ω and $\varphi_\alpha \rightarrow v$ a.e. on V . Of course, this means that $v = 0$ a.e. on $V \setminus \Omega$ and $u = v$ a.e. on Ω . By the same logic as in (1.3), the embedding follows. $\square$

In this proof, we have used the property that $f_m \rightarrow f$ in $L^p(\Omega, d\nu)$ implies the existence of a subsequence (f_{m_k}) converging pointwise almost everywhere to f on Ω , for any measure ν . Consequently, this property must hold for convergent sequences in $W^{1,p}(\Omega, \omega)$:

PROPOSITION 1.3. *Let $\Omega \subseteq \mathbb{R}^n$ be open and fix $1 \leq p < \infty$. Assume that (u_m) is a sequence in $W^{1,p}(\Omega, \omega)$ converging strongly to $u \in W^{1,p}(\Omega, \omega)$. Then, there is a subsequence (u_{m_k}) converging pointwise a.e. to u on Ω .*

As is special with Sobolev spaces, one can sometimes deduce the almost everywhere convergence (up to a subsequence) of the gradients from mere pointwise convergence and boundedness. This continues to be the case in weighted Sobolev spaces as is demonstrated below. Recall that ω continues to denote a lower Lebesgue weight on Ω .

LEMMA 1.4 (Gradient Lifting). *Let Ω be open and bounded with $p \in (1, \infty)$. Let ω be Lebesgue dominating and define $T : \mathbb{R} \rightarrow \mathbb{R}$ by*

$$T(s) := \begin{cases} s & \text{if } |s| \leq 1, \\ \frac{s}{|s|} & \text{if } |s| > 1. \end{cases}$$

Finally, assume that

- (i) $v_\alpha \rightarrow v$ pointwise a.e. on Ω ;
- (ii) (v_α) is bounded in $W^{1,p}(\Omega, \omega)$ and

$$\lim_{\alpha \rightarrow \infty} \int_{\Omega} (|\nabla v_\alpha|^{p-2} \nabla v_\alpha - |\nabla v|^{p-2} \nabla v) \cdot \nabla T(v_\alpha - v) \omega \, dx = 0.$$

Then, after passing to a subsequence,

- (1) $\nabla v_\alpha \rightarrow \nabla v$ pointwise almost everywhere in Ω ;
- (2) $\lim_{n \rightarrow \infty} \left(\|\nabla v_\alpha\|_{L^p(\Omega, \omega)}^p - \|\nabla(v_\alpha - v)\|_{L^p(\Omega, \omega)}^p \right) = \|\nabla v\|_{L^p(\Omega, \omega)}^p$; and
- (3) $|\nabla v_\alpha|^{p-2} \nabla v_\alpha - |\nabla(v_\alpha - v)|^{p-2} \nabla(v_\alpha - v) \rightarrow |\nabla v|^{p-2} \nabla v$ in $L^{\frac{p}{p-1}}(\Omega, \omega)$.

PROOF. Define $d\mu := \omega \, dx$ and note that $\mu(\Omega) < \infty$. Now, $v_n \in W^{1,p}(\Omega)$ and since T is globally Lipschitz, $T(v_n) \in W^{1,p}(\Omega)$. In fact, one has

$$\nabla T(u) = T'(u) \nabla u \quad \text{in } \Omega$$

for each $u \in W^{1,p}(\Omega)$. For each $\alpha \geq 1$,

$$\int_{\Omega} (|\nabla v_\alpha|^{p-2} \nabla v_\alpha - |\nabla v|^{p-2} \nabla v) \cdot \nabla T(v_\alpha - v) \, d\mu \tag{1.4}$$

is equal to

$$\int_{|v_\alpha - v| \leq 1} (|\nabla v_\alpha|^{p-2} \nabla v_\alpha - |\nabla v|^{p-2} \nabla v) \cdot (\nabla v_\alpha - \nabla v) \, d\mu.$$

Or, rather,

$$\begin{aligned} & \int_{\Omega} (|\nabla v_\alpha|^{p-2} \nabla v_\alpha - |\nabla v|^{p-2} \nabla v) \cdot (\nabla v_n - \nabla v) \, d\mu \\ & - \int_{|v_\alpha - v| > 1} (|\nabla v_\alpha|^{p-2} \nabla v_\alpha - |\nabla v|^{p-2} \nabla v) \cdot (\nabla v_\alpha - \nabla v) \, d\mu \end{aligned}$$

Since (∇v_α) and ∇v are bounded in $L^p(\Omega, d\mu)$ and $v_\alpha \rightarrow v$ in the measure μ on Ω , this second term also converges to 0. It follows that

$$\int_{\Omega} (|\nabla v_\alpha|^{p-2} \nabla v_\alpha - |\nabla v|^{p-2} \nabla v) \cdot (\nabla v_\alpha - \nabla v) d\mu \rightarrow 0.$$

As the integrand is non-negative, there then is a subsequence such that $\nabla v_\alpha \rightarrow \nabla v$ μ -a.e. on Ω . Since μ and m share the same null sets, this establishes (1). From here, (2) follows from the Brézis-Lieb lemma. $\square$

2. Homogeneous Sobolev Spaces

We begin with some preliminary set up. Let $\Omega \subset \mathbb{R}^n$ be a non-empty bounded domain, containing the origin, such that $\partial\Omega$ is smooth. Define q by

$$q := \frac{np}{n - p(1 + a - b)}$$

where

$$1 < p < n, \quad a < \frac{n-p}{p}, \quad \text{and} \quad a \leq b < a+1. \quad (2.1)$$

It can be mechanically verified that $qb < n$ and $ap < n$. We will be interested in the following critical equation of the Caffarelli-Kohn-Nirenberg type:

$$\begin{cases} -\operatorname{div}(|x|^{-ap} |\nabla u|^{p-2} \nabla u) = |x|^{-bq} |u|^{q-2} u & \text{in } \Omega, \\ u \in W_0^{1,p}(\Omega, |x|^{-ap}). \end{cases} \quad (2.2)$$

It should be noted here that both the weights $|x|^{-ap}$ and $|x|^{-bq}$ are lower Lebesgue on the whole of $\mathbb{R}^n$. In fact, on Ω , these weights are *Lebesgue dominating*. Hence, the space $W_0^{1,p}(\Omega, |x|^{-ap})$ is well defined.

We also recall the Caffarelli-Kohn-Nirenberg (CKN) inequality which asserts that, for a suitable constant $C = C(n, a, b, p) > 0$, one has

$$\left(\int_{\mathbb{R}^n} |x|^{-bq} |u(x)|^q dx \right)^{1/q} \leq C \left(\int_{\mathbb{R}^n} |x|^{-ap} |\nabla u(x)|^p dx \right)^{1/p} \quad (\text{CKN})$$

for every $u \in C_c^\infty(\mathbb{R}^n)$. This embedding is precisely what motivates the following definition of a Homogeneous space:

DEFINITION 2. Under conditions (2.1), we define $\mathcal{D}^{1,p}(\mathbb{R}^n, |x|^{-ap})$ to be precisely those functions $u \in L^{q^*}(\mathbb{R}^n, |x|^{-bq})$ having the property that $\nabla u \in L^p(\mathbb{R}^n, |x|^{-ap})$. In particular, $\mathcal{D}^{1,p}(\mathbb{R}^n, |x|^{-ap}) \subseteq L_{\text{loc}}^1(\mathbb{R}^n, |x|^{-ap})$. We endow $\mathcal{D}^{1,p}(\mathbb{R}^n)$ with the norm

$$\|u\| := \|\nabla u\|_{L^p(\Omega, |x|^{-ap})}.$$

Finally, we denote by $\mathcal{D}_0^{1,p}(\mathbb{R}^n, |x|^{-ap})$ the closure of $C_c^\infty(\mathbb{R}^n)$ in $\mathcal{D}^{1,p}(\mathbb{R}^n, |x|^{-ap})$.

Our first claim is that $\mathcal{D}_0^{1,p}(\mathbb{R}^n, |x|^{-ap})$ is a Banach space. Since I suspect that there holds $\mathcal{D}_0^{1,p}(\mathbb{R}^n, |x|^{-ap}) = \mathcal{D}^{1,p}(\mathbb{R}^n, |x|^{-ap})$, this may also mean that the latter is Banach.

THEOREM 2.1. *Given conditions (2.1), $\mathcal{D}_0^{1,p}(\mathbb{R}^n, |x|^{-ap})$ is a Banach space. Furthermore, the inequality (CKN) holds for all $u \in \mathcal{D}_0^{1,p}(\mathbb{R}^n, |x|^{-ap})$.*

PROOF. It is best to first verify (CKN) for $\mathcal{D}_0^{1,p}(\mathbb{R}^n, |x|^{-ap})$. Given u in this space, we may select a sequence (φ_α) in $C_c^\infty(\mathbb{R}^n)$ converging to u in $\mathcal{D}_0^{1,p}(\mathbb{R}^n, |x|^{-ap})$. This means that (φ_α) will be Cauchy in $L^q(\mathbb{R}^n, |x|^{-bq})$. Put otherwise, there is $v \in L^q(\mathbb{R}^n, |x|^{-bq})$ such that $\varphi_\alpha \rightarrow v$ in $L^q(\mathbb{R}^n, |x|^{-ap})$. Fix now a test function $\phi \in C_c^\infty(\mathbb{R}^n)$ and set $K := \text{supp}(\phi)$. Since K is compact and $|x|^{-bq}$ is lower Lebesgue, it follows that

$$\begin{aligned} \int_{\mathbb{R}^n} v \partial_i \phi \, dx &= \lim_{\alpha \rightarrow \infty} \int_{\mathbb{R}^n} \varphi_\alpha \partial_i \phi \, dx = - \lim_{\alpha \rightarrow \infty} \int_{\mathbb{R}^n} \phi \partial_i \varphi_\alpha \, dx \\ &= - \int_{\mathbb{R}^n} \phi \partial_i u \, dx. \end{aligned}$$

In this last step, we have used the fact that $|x|^{-ap}$ is lower Lebesgue to ensure that $\nabla \varphi_\alpha \rightarrow \nabla u$ in $L^p(K, |x|^{-ap})$ implies this same convergence in $L^p(K)$. From here, we infer that $\nabla v = \nabla u$ in the weak sense whence $v - u$ is constant. However, since $bq < n$ and $v - u \in L^q(\mathbb{R}^n, |x|^{-bq})$, we must have $v - u \equiv 0$. In short, $\varphi_\alpha \rightarrow u$ in $L^q(\mathbb{R}^n, |x|^{-bq})$ whence (CKN) holds for u by continuity of norms.

Next we prove that $\mathcal{D}_0^{1,p}(\mathbb{R}^n, |x|^{-ap})$ is a Banach space. Fix a Cauchy sequence (u_α) and note that the sequences (u_α) and (∇u_α) are Cauchy in $L^q(\mathbb{R}^n, |x|^{-bq})$ and $L^p(\mathbb{R}^n, |x|^{-ap})$, respectively. Let $u := \lim u_\alpha$ and $u_i := \lim \partial_i u_\alpha$. For $\phi \in C_c^\infty(\mathbb{R}^n)$, the lower Lebesgue property guarantees that

$$\begin{aligned} \int_{\mathbb{R}^n} v \partial_i \phi \, dx &= \lim_{\alpha \rightarrow \infty} \int_{\mathbb{R}^n} u_\alpha \partial_i \phi \, dx = - \lim_{\alpha \rightarrow \infty} \int_{\mathbb{R}^n} \phi \partial_i u_\alpha \, dx \\ &= - \int_{\mathbb{R}^n} \phi u_i \, dx \end{aligned}$$

for all $i = 1, \dots, n$. Hence, $u \in \mathcal{D}_0^{1,p}(\mathbb{R}^n, |x|^{-ap})$ and $u_\alpha \rightarrow u$. $\square$

2.1. A Rellich-Kondrachov Type Embedding Theorem. The (CKN) inequality, which is valid for all $u \in \mathcal{D}_0^{1,p}(\mathbb{R}^n, |x|^{-ap})$, also holds true in $W_0^{1,p}(\Omega, |x|^{-ap})$. Indeed, any $u \in W_0^{1,p}(\Omega, |x|^{-ap})$ can be approximated by (φ_α) in $C_c^\infty(\Omega)$ and hence, after passing to a subsequence, $\varphi_\alpha \rightarrow u$ a.e. on Ω . However, being Cauchy in $W_0^{1,p}(\Omega, |x|^{-ap})$ implies by (CKN) that (φ_α) is Cauchy in $L^q(\Omega, |x|^{-bq})$ with limit v in

this same space. Since there is a further subsequence converging almost everywhere to v on Ω , we must have $v = u$. Therefore,

$$\begin{aligned} \|u\|_{L^q(\Omega, |x|^{-bq})} &= \lim_{\alpha \rightarrow \infty} \|\varphi_\alpha\|_{L^q(\Omega, |x|^{-bq})} \leq C \lim_{\alpha \rightarrow \infty} \|\nabla \varphi_\alpha\|_{L^p(\Omega, |x|^{-ap})} \\ &= C \|\nabla u\|_{L^p(\Omega, |x|^{-ap})} \\ &\leq C \|u\|_{W^{1,p}(\Omega, |x|^{-ap})}. \end{aligned}$$

Thus, the identity operator is an embedding

$$W_0^{1,p}(\Omega, |x|^{-ap}) \hookrightarrow L^q(\Omega, |x|^{-bq}).$$

Since the measure $|x|^{-bq} dx$ is finite on compact sets, we obtain through composition and embedding

$$W_0^{1,p}(\Omega, |x|^{-ap}) \hookrightarrow L^r(\Omega, |x|^{-bq}).$$

whenever $1 \leq r < q$. In fact, this gives the embedding $W_0^{1,p}(\Omega, |x|^{-ap}) \hookrightarrow L^r(\Omega, |x|^{-\theta})$ for all $\theta \leq bq$. Next, we ask ourselves when this embedding is compact:

THEOREM 2.2. *Let $1 \leq r < q$ and $-\infty < \theta \leq bq$. Then, the embedding*

$$W_0^{1,p}(\Omega, |x|^{-ap}) \hookrightarrow L^r(\Omega, |x|^{-\theta})$$

is compact.

PROOF. Since $|x|^{-\theta} \lesssim |x|^{-bq}$ on Ω , it is enough to prove the theorem for $\theta = bq$. Let (u_α) be a bounded sequence in $W_0^{1,p}(\mathbb{R}^n, |x|^{-ap})$. By our previous observations, this sequence is bounded in $L^q(\Omega, |x|^{-bq})$. Furthermore, it is easy to see that (u_α) is bounded in $W^{1,p}(\Omega)$ whence it has (by the classical Rellich-Kondrachov theorem) a subsequence (also denoted (u_α)) converging strongly to some $u \in L^p(\Omega)$, and almost everywhere on Ω . Now, it is clear from Fatou's lemma that $u \in L^q(\Omega, |x|^{-bq})$ and so $\{|u_\alpha - u|^r\}_{\alpha \in \mathbb{N}}$ is uniformly bounded in $L^{q/r}(\Omega, |x|^{-bq})$, where $q/r > 1$. Finally, using that $|x|^{-bq} dx$ is finite on Ω , the p -bounded convergence theorem implies that

$$\lim_{\alpha \rightarrow \infty} \int_{\Omega} |u_\alpha - u|^r |x|^{-bq} dx = \lim_{\alpha \rightarrow \infty} \int_{\Omega} |u_\alpha - u|^r |x|^{-bq} \mathbf{1}_{\Omega} dx = 0$$

which completes the proof. $\square$